

STAT 314: APPLIED DYNAMICAL SYSTEM
FALL 2025
PROBLEM SET 5

JUNHO LEE
JHLEE99@UCHICAGO.EDU

1.

Solution. (p.187, Problem 10) Consider the system:

$$\dot{x} = y + 2z + (x + z)^2 + xy - y^2, \quad \dot{y} = (x + z)^2, \quad \dot{z} = -2z - (x + z)^2 + y^2.$$

(a) Separating its linear and nonlinear parts, we have

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} (x + z)^2 + xy - y^2 \\ (x + z)^2 \\ -(x + z)^2 + y^2 \end{pmatrix}.$$

Let this linear part be represented by the matrix A :

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

The characteristic polynomial of A is given by

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 & 2 \\ 0 & -\lambda & 0 \\ 0 & 0 & -2 - \lambda \end{pmatrix} = -\lambda^2(-2 - \lambda) = 0.$$

Thus, the eigenvalues are $\lambda_1 = 0$ (with algebraic multiplicity 2) and $\lambda_2 = -2$.

Now, we find the eigenvectors. For $\lambda_1 = 0$, we trivially have $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. To find a generalized eigenvector, we solve $Av_2 = v_1$:

$$\begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

This gives us $v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. For $\lambda_2 = -2$, we solve $(A + 2I)v = 0$:

$$\begin{pmatrix} 2 & 1 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

This gives us $v_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

Hence we obtain relation between old and new coordinates:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} \implies \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} x + z \\ y \\ z \end{pmatrix}.$$

Thus, let $u = x + z, v = y, w = z$, where (u, v) corresponds to the center subspace and w corresponds to the stable subspace. Here we have

$$\begin{aligned} \dot{u} &= \dot{x} + \dot{z} = y + xy = y + (x + z)y - zy = v + uv - vw \\ \dot{v} &= (x + z)^2 = u^2 \\ \dot{w} &= -2z - (x + z)^2 + y^2 = -2w - u^2 + v^2. \end{aligned}$$

rewriting the system in terms of u, v, w , we have

$$\begin{pmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} + \begin{pmatrix} uv - vw \\ u^2 \\ v^2 - u^2 \end{pmatrix}.$$

and hence

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = C \begin{bmatrix} u \\ v \end{bmatrix} + F(u, v, w), \quad \dot{w} = Sw + G(u, v, w),$$

where $C = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $S = -2$ and $F(u, v, w) = \begin{pmatrix} uv - vw \\ u^2 \end{pmatrix}$, $G(u, v, w) = v^2 - u^2$.

(b) To find the center manifold, we assume it is of the form $w = h(u, v)$ with $h(0, 0) = 0$ and $Dh(0, 0) = 0$. We approximate $h(u, v)$ by a polynomial up to second order:

$$h(u, v) = au^2 + buv + cv^2 + o((u^2 + v^2)).$$

By plugging $w = h(u, v)$ into the equation for $\dot{w}$ and using the chain rule, we have

$$\frac{\partial h}{\partial u} \dot{u} + \frac{\partial h}{\partial v} \dot{v} = Sh(u, v) + G(u, v, h(u, v)).$$

Calculating the left-hand side, we get

$$\frac{\partial h}{\partial u} = 2au + bv, \quad \frac{\partial h}{\partial v} = bu + 2cv.$$

Thus,

$$\begin{aligned} \frac{\partial h}{\partial u} \dot{u} + \frac{\partial h}{\partial v} \dot{v} &= (2au + bv)(v + uv - h(u, v)v) + (bu + 2cv)(u^2) + o((u^2 + v^2)) \\ &= (2au + bv)(v + uv - (au^2 + buv + cv^2)v) + (bu + 2cv)(u^2) + o((u^2 + v^2)) \\ &= (2au + bv)(v + uv - au^2v - buv^2 - cv^3) + (bu + 2cv)(u^2) + o((u^2 + v^2)) \\ &= 2auv + bv^2 + o((u^2 + v^2)) \end{aligned}$$

Calculating the right-hand side, we have

$$Sh(u, v) + G(u, v, h(u, v)) = -2(au^2 + buv + cv^2) + v^2 - u^2 + o((u^2 + v^2))$$

Hence equating both sides, we get

$$2auv + bv^2 = -2au^2 - 2buv - 2cv^2 + v^2 - u^2.$$

Matching coefficients, we have

$$\begin{cases} -2a = 1 \\ -2b - 2a = 0 \\ -2c + 1 = b \end{cases} \implies \begin{cases} a = -\frac{1}{2} \\ b = \frac{1}{2} \\ c = \frac{1}{4} \end{cases}.$$

Thus, the center manifold is approximated by

$$h(u, v) = -\frac{1}{2}u^2 + \frac{1}{2}uv + \frac{1}{4}v^2 + o((u^2 + v^2)).$$

(c) On the center manifold, we have $w = h(u, v)$. Thus, the reduced system on the center manifold is given by

$$\begin{aligned} \dot{u} &= v + uv - h(u, v)v = v + uv - \left(-\frac{1}{2}u^2 + \frac{1}{2}uv + \frac{1}{4}v^2\right)v \\ &= v + uv + \frac{1}{2}u^2v - \frac{1}{2}uv^2 - \frac{1}{4}v^3 \\ \dot{v} &= u^2. \end{aligned}$$

To visualize the dynamics on the center manifold, we can plot the phase portrait of the reduced system up to cubic terms using Python:

```
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt

# Symbols
u, v = sp.symbols('u v', real=True)

# Center manifold approximation: w = h(u, v)
a = -sp.Rational(1, 2)
b = sp.Rational(1, 2)
c = sp.Rational(1, 4)
h = a*u**2 + b*u*v + c*v**2

# Reduced dynamics on the center manifold (up to cubic terms)
du = v + u*v - h*v      # u' = v + uv - h(u,v)*v
dv = u**2                 # v' = u^2

# Expand for clarity (optional)
du_simplified = sp.expand(du)
dv_simplified = sp.expand(dv)
print("du =", du_simplified)
print("dv =", dv_simplified)

# Turn into numerical functions
du_fun = sp.lambdify((u, v), du_simplified, 'numpy')
dv_fun = sp.lambdify((u, v), dv_simplified, 'numpy')

# Grid for plotting
U, V = np.meshgrid(
    np.linspace(-0.8, 0.8, 25),
    np.linspace(-0.8, 0.8, 25)
)

DU = du_fun(U, V)
```

```

DV = dv_fun(U, V)

plt.figure(figsize=(5, 4))
plt.streamplot(U, V, DU, DV, density=1.2, linewidth=1, arrowsize=1.2)

# Axes / labels
plt.axhline(0, linewidth=0.5)
plt.axvline(0, linewidth=0.5)
plt.xlabel('u')
plt.ylabel('v')
plt.title('Reduced dynamics on the center manifold (up to cubic terms)')
plt.gca().set_aspect('equal', 'box')
plt.tight_layout()
plt.show()

```

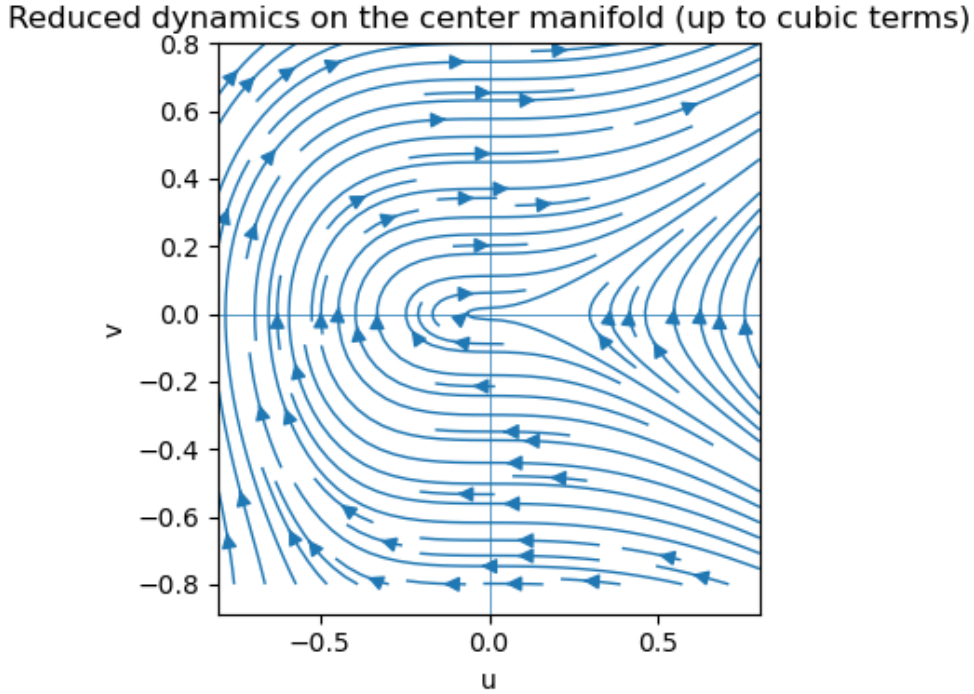


FIGURE 1. Phase portrait of the reduced system on the center manifold (up to cubic terms).

Roughly speaking, the reduced system satisfies

$$\dot{u} \approx v, \quad \dot{v} \approx u^2.$$

Thus, near the origin, u increases when $v > 0$ and decreases when $v < 0$, while $\dot{v} \geq 0$ with $\dot{v} > 0$ whenever $u \neq 0$. In particular, any trajectory with $u(0) \neq 0$ has $v(t)$ strictly increasing for small positive times. Trajectories starting below the u -axis ($v < 0$) therefore cross the axis in finite time and then move upward, and those starting above the axis continue to move upward.

Consequently, for any neighborhood of the origin there exist initial conditions in that neighborhood whose trajectories leave it in forward time. Hence the origin is a (nonlinearly) unstable equilibrium on the center manifold. The phase portrait in Figure 1 illustrates this behavior, with trajectories moving away from the origin as t increases.

Solution. (p.314-316, Problem 8) Let $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, consider $\mathcal{H}_2^2, \mathcal{H}_3^2$ as the spaces of homogeneous polynomials of degree 2 and 3 in two variables, respectively, i.e.

$$\mathcal{H}_2^2 = \text{span} \left\{ \begin{pmatrix} x^2 \\ 0 \end{pmatrix}, \begin{pmatrix} xy \\ 0 \end{pmatrix}, \begin{pmatrix} y^2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x^2 \end{pmatrix}, \begin{pmatrix} 0 \\ xy \end{pmatrix}, \begin{pmatrix} 0 \\ y^2 \end{pmatrix} \right\},$$

and

$$\mathcal{H}_3^2 = \text{span} \left\{ \begin{pmatrix} x^3 \\ 0 \end{pmatrix}, \begin{pmatrix} x^2y \\ 0 \end{pmatrix}, \begin{pmatrix} xy^2 \\ 0 \end{pmatrix}, \begin{pmatrix} y^3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x^3 \end{pmatrix}, \begin{pmatrix} 0 \\ x^2y \end{pmatrix}, \begin{pmatrix} 0 \\ xy^2 \end{pmatrix}, \begin{pmatrix} 0 \\ y^3 \end{pmatrix} \right\}.$$

Given a homological operator $L_A : \mathcal{H}_n^2 \rightarrow \mathcal{H}_n^2$ defined by

$$L_A(h)(x, y) = Ah(x, y) - Dh(x, y)A \begin{pmatrix} x \\ y \end{pmatrix},$$

we want to find the matrix representation of L_A with respect to the bases of $\mathcal{H}_2^2$ and $\mathcal{H}_3^2$.

First, we compute L_A on the basis elements of $\mathcal{H}_2^2$:

$$\begin{aligned} L_A \begin{pmatrix} x^2 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ x^2 \end{pmatrix} - \begin{pmatrix} 2x & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} 2xy \\ x^2 \end{pmatrix}, \\ L_A \begin{pmatrix} xy \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ xy \end{pmatrix} - \begin{pmatrix} y & x \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} x^2 - y^2 \\ xy \end{pmatrix}, \\ L_A \begin{pmatrix} y^2 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ y^2 \end{pmatrix} - \begin{pmatrix} 0 & 2y \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} 2xy \\ y^2 \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ x^2 \end{pmatrix} &= \begin{pmatrix} -x^2 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 2x & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -x^2 \\ 2xy \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ xy \end{pmatrix} &= \begin{pmatrix} -xy \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ y & x \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -xy \\ y^2 - x^2 \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ y^2 \end{pmatrix} &= \begin{pmatrix} -y^2 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 2y \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -y^2 \\ -2xy \end{pmatrix}. \end{aligned}$$

Thus, the matrix representation of L_A on $\mathcal{H}_2^2$ with respect to the given basis is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 & 0 \end{pmatrix}.$$

The eigenvalues of this matrix are $\lambda = -3i, -i, i, 3i$ with corresponding eigenvectors as follows:

- $\lambda = -3i$ (Algebraic mult. 1, Geometric mult. 1)

$$\begin{pmatrix} ix^2 - 2xy - iy^2 \\ -x^2 - 2ixy + y^2 \end{pmatrix}$$

- $\lambda = -i$ (Algebraic mult. 2, Geometric mult. 2)

$$\begin{pmatrix} y(ix - y) \\ x(-ix + y) \end{pmatrix}, \quad \begin{pmatrix} i(-x^2 - y^2) \\ x^2 + y^2 \end{pmatrix}$$

- $\lambda = i$ (Algebraic mult. 2, Geometric mult. 2)

$$\begin{pmatrix} y(-ix - y) \\ x(ix + y) \end{pmatrix}, \quad \begin{pmatrix} i(x^2 + y^2) \\ x^2 + y^2 \end{pmatrix}$$

- $\lambda = 3i$ (Algebraic mult. 1, Geometric mult. 1)

$$\begin{pmatrix} -ix^2 - 2xy + iy^2 \\ -x^2 + 2ixy + y^2 \end{pmatrix}$$

Since all eigenvalues are non-zero, the operator L_A is invertible on $\mathcal{H}_2^2$, and there are no resonant terms at the quadratic level.

Next, we compute L_A on the basis elements of $\mathcal{H}_3^2$:

$$\begin{aligned} L_A \begin{pmatrix} x^3 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ x^3 \end{pmatrix} - \begin{pmatrix} 3x^2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} 3x^2y \\ x^3 \end{pmatrix}, \\ L_A \begin{pmatrix} x^2y \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ x^2y \end{pmatrix} - \begin{pmatrix} 2xy & x^2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} x^3 + xy^2 \\ x^2y \end{pmatrix}, \\ L_A \begin{pmatrix} xy^2 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ xy^2 \end{pmatrix} - \begin{pmatrix} y^2 & 2xy \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} 2x^2y + y^3 \\ xy^2 \end{pmatrix}, \\ L_A \begin{pmatrix} y^3 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ y^3 \end{pmatrix} - \begin{pmatrix} 0 & 3y^2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} 3xy^2 \\ y^3 \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ x^3 \end{pmatrix} &= \begin{pmatrix} -x^3 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 3x^2 & 0 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -x^3 \\ 3x^2y \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ x^2y \end{pmatrix} &= \begin{pmatrix} -x^2y \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 2xy & x^2 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -x^2y \\ x^3 + xy^2 \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ xy^2 \end{pmatrix} &= \begin{pmatrix} -xy^2 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ y^2 & 2xy \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -xy^2 \\ 2x^2y + y^3 \end{pmatrix}, \\ L_A \begin{pmatrix} 0 \\ y^3 \end{pmatrix} &= \begin{pmatrix} -y^3 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 3y^2 \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} -y^3 \\ 3xy^2 \end{pmatrix}. \end{aligned}$$

Thus, the matrix representation of L_A on $\mathcal{H}_3^2$ with respect to the given basis is

$$\begin{pmatrix} 0 & 1 & 2 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -2 & -3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \end{pmatrix}.$$

- $\lambda = 0$ (Algebraic mult. 2, Geometric mult. 2)

These eigenvectors span the **Kernel (Right Nullspace)** of L_3 :

$$\begin{pmatrix} -y(x^2 + y^2) \\ x(x^2 + y^2) \end{pmatrix}, \quad \begin{pmatrix} x(x^2 + y^2) \\ y(x^2 + y^2) \end{pmatrix}$$

- $\lambda = -4i$ (Algebraic mult. 1, Geometric mult. 1)

$$\begin{pmatrix} x^3 + 3ix^2y - 3xy^2 - iy^3 \\ ix^3 - 3x^2y - 3ixy^2 + y^3 \end{pmatrix}$$

- $\lambda = -2i$ (Algebraic mult. 2, Geometric mult. 2)

$$\begin{pmatrix} y(x^2 + 2ixy - y^2) \\ x(-x^2 - 2ixy + y^2) \end{pmatrix}, \quad \begin{pmatrix} -x^3 - 3xy^2 - 2iy^3 \\ -2ix^3 + 3x^2y + y^3 \end{pmatrix}$$

- $\lambda = 2i$ (Algebraic mult. 2, Geometric mult. 2)

$$\begin{pmatrix} y(x^2 - 2ixy - y^2) \\ x(-x^2 + 2ixy + y^2) \end{pmatrix}, \quad \begin{pmatrix} -x^3 - 3xy^2 + 2iy^3 \\ 2ix^3 + 3x^2y + y^3 \end{pmatrix}$$

- $\lambda = 4i$ (Algebraic mult. 1, Geometric mult. 1)

$$\begin{pmatrix} x^3 - 3ix^2y - 3xy^2 + iy^3 \\ -ix^3 - 3x^2y + 3ixy^2 + y^3 \end{pmatrix}$$

The coordinate vector for $\mathbf{v}_1$ is $\mathbf{c}_1 = (3, 0, 1, 0, 0, 1, 0, 3)^T$. We calculate the product $L_3^T \mathbf{c}_1$:

$$\begin{aligned} L_3^T \mathbf{c}_1 &= \begin{pmatrix} 0 & 3 & 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 & 0 & -3 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 0(3) + 0(1) + 1(0) + 0(3) \\ -1(3) + 2(1) + 1(1) + 0(3) \\ 0(3) + 0(1) + 0(1) + 0(3) \\ 0(3) - 3(1) + 0(1) + 1(3) \\ -1(3) + 0(1) + 3(1) + 0(3) \\ 0(3) + 0(1) - 1(0) + 0(3) \\ 0(3) - 1(1) - 2(1) + 1(3) \\ 0(3) + 0(1) + 0(1) + 0(3) \end{pmatrix} = \begin{pmatrix} 0 \\ -3 + 2 + 1 \\ 0 \\ -3 + 3 \\ -3 + 3 \\ 0 \\ -1 - 2 + 3 \\ 0 \end{pmatrix} = \mathbf{0}. \end{aligned}$$

The coordinate vector for $\mathbf{v}_2$ is $\mathbf{c}_2 = (0, 1, 0, 3, -3, 0, -1, 0)^T$. We calculate the product $L_3^T \mathbf{c}_2$:

$$\begin{aligned} L_3^T \mathbf{c}_2 &= \begin{pmatrix} 0 & 3 & 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 & 0 & -3 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 3 \\ -3 \\ 0 \\ -1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 3(1) + 1(-3) \\ 0 \\ -2(1) + 1(3) + 1(-1) \\ 0 \\ 0 \\ -1(1) - 1(-3) + 2(-1) \\ 0 \\ -1(3) - 3(-1) \end{pmatrix} = \begin{pmatrix} 3 - 3 \\ 0 \\ -2 + 3 - 1 \\ 0 \\ 0 \\ -1 + 3 - 2 \\ 0 \\ -3 + 3 \end{pmatrix} = \mathbf{0}. \end{aligned}$$

which shows that both $\mathbf{v}_1$ and $\mathbf{v}_2$ lie in the kernel of the adjoint operator L_A^* .

Finally, for the homological operator L_3 acting on $\mathcal{H}_3^2$, the spectral computation gives

$$\text{alg. mult}(0) = 2, \quad \text{geom. mult}(0) = 2.$$

This means that the eigenvalue 0 has exactly two linearly independent eigenvectors and no additional generalized eigenvectors. In this situation every vector v satisfying $Lv = 0$ is a

genuine eigenvector, and no nonzero vector in $\ker L$ can lie in $\text{range } L$: if $v = Lw$ and $Lv = 0$, then w itself would have to be an eigenvector, which would force $Lw = 0$, contradicting $v \neq 0$. Therefore,

$$\text{range } L_3 \cap \ker L_3 = \{0\}.$$

As a result we obtain the direct decomposition

$$\mathcal{H}_3^2 = \text{range } L_3 \oplus \ker L_3,$$

and the two computed right-null vectors form a basis of $\ker L_3$, while together with $\text{range } L_3$ they span the whole space $\mathcal{H}_3^2$.

Solution. (p.314-316, Problem 13) (a) Let

$$\begin{aligned}\dot{x} &= \mu x - y + ay^2 + x^3 \\ \dot{y} &= x + \mu y + xy^2 + y^2,\end{aligned}$$

Then its equilibrium point is at the origin $(x, y) = (0, 0)$. By separating the linear and nonlinear parts, we have

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} ay^2 + x^3 \\ xy^2 + y^2 \end{pmatrix}.$$

and its Jacobian at the origin is

$$J = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}.$$

The characteristic polynomial of J is given by

$$\det(J - \lambda I) = \det \begin{pmatrix} \mu - \lambda & -1 \\ 1 & \mu - \lambda \end{pmatrix} = (\mu - \lambda)^2 + 1 = 0.$$

Thus, the eigenvalues are $\lambda = \mu \pm i$. When $\mu = 0$, the eigenvalues are purely imaginary, $\lambda = \pm i$.

By plugging $\mu = 0$ into the system, we have

$$\begin{aligned}\dot{x} &= -y + ay^2 + x^3 \\ \dot{y} &= x + xy^2 + y^2,\end{aligned}$$

hence by letting $p(x, y) = ay^2 + x^3$ and $q(x, y) = xy^2 + y^2$, we can apply Guckenheimer's theorem to obtain

$$\begin{aligned}\alpha(a) &= \frac{1}{16} (p_{xxx} + p_{xyy} + q_{xxy} + q_{yyy}) \\ &\quad + \frac{1}{16\omega} (p_{xy}(p_{xx} + p_{yy}) - q_{xy}(q_{xx} + q_{yy}) - p_{xx}q_{xx} + p_{yy}q_{yy}),\end{aligned}$$

where $\omega = 1$. Now,

$$\frac{1}{16} (p_{xxx} + p_{xyy} + q_{xxy} + q_{yyy}) = \frac{1}{16} (6 + 0 + 0 + 0) = \frac{6}{16} = \frac{3}{8},$$

and $q_{xx} = q_{xy} = p_{xx} = 0$ at $(0, 0)$, $p_{yy} = 2a$, $q_{yy} = 2$, so

$$\frac{1}{16\omega} (p_{xy}(p_{xx} + p_{yy}) - q_{xy}(q_{xx} + q_{yy}) - p_{xx}q_{xx} + p_{yy}q_{yy}) = \frac{1}{16} (0 - 0 - 0 + 4a) = \frac{4a}{16} = \frac{a}{4}.$$

Thus, we have

$$\alpha(a) = \frac{3}{8} + \frac{a}{4} = \frac{3 + 2a}{8}.$$

(b) The linearization at the origin has eigenvalues

$$\lambda(\mu) = \mu \pm i,$$

so a Hopf bifurcation can only occur when the real part vanishes, i.e.

$$\mu = 0.$$

Here $\frac{d}{d\mu} \Re \lambda_{1,2}(\mu)|_{\mu=0} = 1 \neq 0$, and the transversality condition for a Hopf bifurcation is automatically satisfied.

From part (a) we have computed the first Lyapunov coefficient

$$\alpha(a) = \frac{3 + 2a}{8}.$$

Thus the system undergoes an Andronov–Hopf bifurcation at $\mu = 0$ whenever $\alpha(a) \neq 0$, i.e. for all

$$a \neq -\frac{3}{2}.$$

The sign of $\alpha(a)$ determines the type of Hopf bifurcation:

$$\begin{cases} \alpha(a) < 0 & \Rightarrow \text{supercritical Hopf (a stable small limit cycle is born),} \\ \alpha(a) > 0 & \Rightarrow \text{subcritical Hopf (an unstable small limit cycle is born).} \end{cases}$$

Since

$$\alpha(a) < 0 \iff a < -\frac{3}{2}, \quad \alpha(a) > 0 \iff a > -\frac{3}{2},$$

At the degenerate value $a = -\frac{3}{2}$, the first Lyapunov coefficient vanishes and the standard Hopf bifurcation theorem does not apply.

(c) On the center manifold the Hopf normal form can be written in complex form as

$$\dot{z} = (\lambda(\mu) + c(\mu)|z|^2 + \mathcal{O}(|z|^4)) z,$$

with

$$\lambda(\mu) = \mu + i\omega + \mathcal{O}(\mu^2), \quad c(\mu) = c_0 + \mathcal{O}(\mu), \quad \alpha(a) := \Re c_0 = \frac{3 + 2a}{8}.$$

Writing $z = re^{i\theta}$ and separating real and imaginary parts gives

$$\dot{r} = r(\mu + \alpha(a)r^2 + \mathcal{O}(\mu r^2, r^4)), \quad \dot{\theta} = \omega + \mathcal{O}(\mu, r^2).$$

A nontrivial periodic orbit corresponds to an equilibrium of the radial equation with $r > 0$, i.e. a solution of

$$\mu + \alpha(a)r^2 + \mathcal{O}(r^4) = 0.$$

For $|\mu|$ sufficiently small and $\mu \alpha(a) < 0$ this has a unique small solution

$$r_*(\mu; a) = \sqrt{-\frac{\mu}{\alpha(a)}} + \mathcal{O}(|\mu|^{3/2}),$$

so, to leading order,

$$r_*(\mu; a) \sim \sqrt{-\mu/\alpha(a)}.$$

Figure 2 illustrates these predictions for representative values of a and μ . Recall that

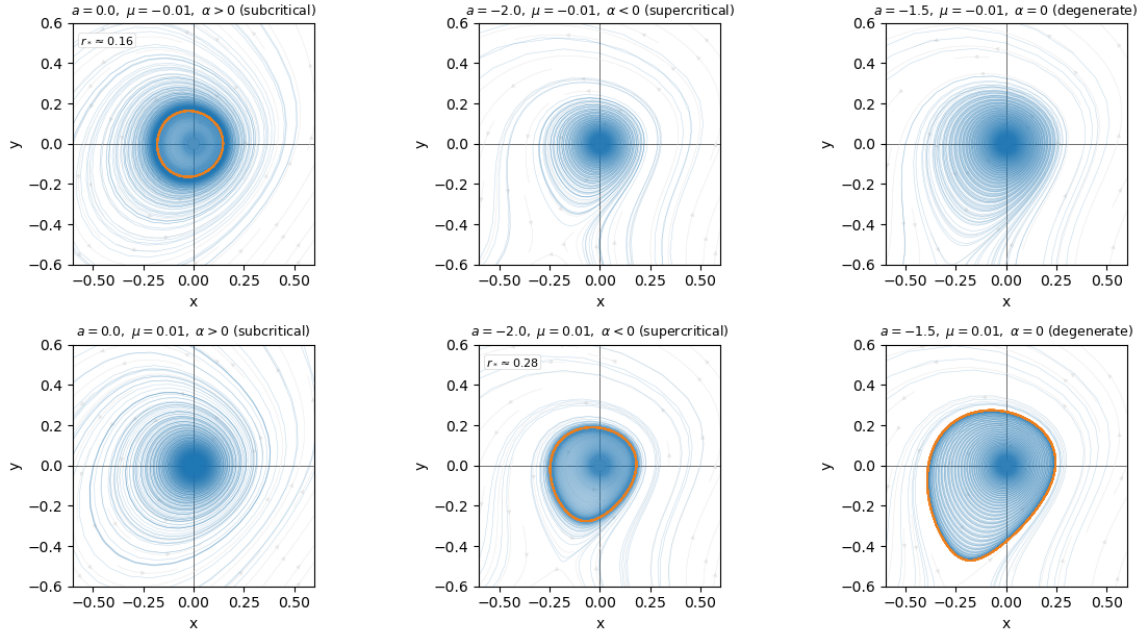
$$\alpha(a) = \frac{3 + 2a}{8},$$

so that, for example, $a = 0$ gives $\alpha > 0$ (subcritical Hopf), $a = -2$ gives $\alpha < 0$ (supercritical Hopf), and $a = -\frac{3}{2}$ corresponds to the degenerate case $\alpha = 0$.

• $\alpha(a) > 0$ (**subcritical Hopf, e.g. $a = 0$**).

- For $\mu < 0$ there is a small *unstable* limit cycle with radius $r_* \approx \sqrt{-\mu/\alpha(a)}$, surrounding a stable focus at the origin. Trajectories inside the cycle spiral to the origin, while trajectories starting just outside spiral away.
- For $\mu > 0$ the small unstable cycle disappears and the origin becomes an unstable focus; no small limit cycle is visible near the Hopf point.

Numerical phase portraits near the Hopf bifurcation

FIGURE 2. Phase portraits for $\alpha < 0, \alpha = 0, \alpha > 0$ and $\mu < 0, \mu > 0$.

- $\alpha(a) < 0$ (**supercritical Hopf**, e.g. $a = -2$).
 - For $\mu < 0$ the origin is a stable focus and there is no small limit cycle.
 - For $\mu > 0$ a unique small *stable* limit cycle is created, with radius $r_* \approx \sqrt{-\mu/\alpha(a)}$. Trajectories starting inside or outside this radius are attracted to the cycle, while the origin becomes unstable.
- $\alpha(a) = 0$ (**degenerate Hopf**, $a = -\frac{3}{2}$).
 - Here the cubic term in the normal form vanishes and higher-order terms determine the dynamics. Analytically the standard Hopf formula for r_* does not apply.
 - Numerically, for $\mu < 0$ we only observe a stable focus at the origin with no small isolated cycle.
 - For $\mu > 0$ we observe a *semi-stable* limit cycle: trajectories starting inside the loop approach this closed orbit, whereas trajectories starting just outside are repelled and move away.

2. Let

$$\begin{aligned}\frac{\partial w}{\partial t} &= p - w - wb^2 + \nu \frac{\partial w}{\partial x}, \\ \frac{\partial b}{\partial t} &= -mb + wb^2 + \frac{\partial^2 b}{\partial x^2},\end{aligned}$$

Solution. (a) $\frac{\partial w}{\partial x} > 0$ implies that the water concentration w increases as x increases, and vice versa. For $\nu > 0$, the term $\nu \frac{\partial w}{\partial x} > 0$ means that, when there is more water slightly to the right (at $x + \delta x$), the water concentration w at x increases with time t . In other words, there is a net transport of water from right to left.

Since water flows from high altitude to low altitude, having water effectively move from right to left means that the terrain must be higher to the right and lower to the left.

More precisely, ignoring the reaction terms, the water equation reduces to

$$w_t = \nu w_x.$$

This can be written as a pure transport equation

$$w_t + cw_x = 0$$

with transport speed $c = -\nu$. For $w_t + cw_x = 0$, solutions have the form

$$w(x, t) = f(x - ct),$$

so the profile moves rigidly with velocity c .

Thus, if $\nu > 0$ then $c = -\nu < 0$, and the water profile moves in the direction of decreasing x , i.e. from right to left. Since water flows downhill, the downhill direction must be from right to left, so the elevation is higher to the right and lower to the left.

- (b) Let $\frac{\partial b}{\partial t} = 0$, $\frac{\partial w}{\partial t} = 0$ and $\frac{\partial w}{\partial t} = 0$, $\frac{\partial w}{\partial x} = 0$. Then, we have

$$0 = p - w - wb^2,$$

$$0 = -mb + wb^2.$$

From the second equation, we have either $b = 0$ or $w = m/b$. If $b = 0$, then from the first equation we have $w = p$. Thus, one equilibrium point is $(w, b) = (p, 0)$. If $w = m/b$, then substituting it into the first equation gives

$$0 = p - \frac{m}{b} - \frac{m}{b}b^2 = p - \frac{m}{b} - mb.$$

Multiplying both sides by b gives

$$0 = pb - m - mb^2.$$

Rearranging for b gives a quadratic equation

$$mb^2 - pb + m = 0.$$

Viewing this as a quadratic in b , the quadratic formula yields

$$b = \frac{p \pm \sqrt{p^2 - 4m^2}}{2m}.$$

Real solutions exist only when $p^2 - 4m^2 \geq 0$, i.e. $p \geq 2m$.

- If $p < 2m$, there is no solution with $b > 0$.

- If $p = 2m$, the discriminant is zero and we obtain a single vegetated equilibrium

$$b = 1, \quad w = \frac{m}{b} = m.$$

- If $p > 2m$, there are two distinct vegetated equilibria with $b > 0$:

$$(w, b) = \left(\frac{m}{b}, b \right) = \left(\frac{2m^2}{p \pm \sqrt{p^2 - 4m^2}}, \frac{p \pm \sqrt{p^2 - 4m^2}}{2m} \right).$$

Now, computing the Jacobian matrix at an equilibrium (w^*, b^*) gives

$$J = \begin{pmatrix} -1 - (b^*)^2 & -2w^*b^* \\ (b^*)^2 & -m + 2w^*b^* \end{pmatrix}.$$

Evaluating the Jacobian at the bare-soil equilibrium $(w^*, b^*) = (p, 0)$ gives

$$J = \begin{pmatrix} -1 & 0 \\ 0 & -m \end{pmatrix},$$

which has eigenvalues $\lambda_1 = -1$ and $\lambda_2 = -m$. Thus, the bare-soil equilibrium is stable for all $p > 0$.

Next, we evaluate the Jacobian at the vegetated equilibria, where $w^* = m/b^*$. This gives

$$J = \begin{pmatrix} -1 - (b^*)^2 & -2m \\ (b^*)^2 & m \end{pmatrix}.$$

The characteristic polynomial is given by

$$\det(J - \lambda I) = \det \begin{pmatrix} -1 - (b^*)^2 - \lambda & -2m \\ (b^*)^2 & m - \lambda \end{pmatrix} = 0.$$

Expanding the determinant gives

$$\begin{aligned} \det(J - \lambda I) &= (-1 - (b^*)^2 - \lambda)(m - \lambda) - (-2m)(b^*)^2 \\ &= \lambda^2 + \lambda(1 + (b^*)^2 - m) - m(1 + (b^*)^2) + 2m(b^*)^2 \\ &= \lambda^2 + \underbrace{(1 + (b^*)^2 - m)}_{-\text{tr}(J)} \lambda + \underbrace{m((b^*)^2 - 1)}_{\det(J)} = 0. \end{aligned}$$

From the equilibrium condition $mb^2 - pb + m = 0$, the product of the two roots is $b_+b_- = 1$. Since $p > 2m$, both roots are positive and satisfy

$$0 < b_- < 1 < b_+.$$

Using these inequalities we determine the stability of each branch:

- For the lower branch b_- , we have $(b_-)^2 < 1$, so $\det(J) = m((b_-)^2 - 1) < 0$. Thus the eigenvalues have opposite signs and b_- is an unstable **saddle**.
- For the upper branch b_+ , we have $(b_+)^2 > 1$, hence $\det(J) > 0$. Moreover

$$\text{tr}(J) = m - (1 + (b_+)^2) < 0$$

automatically holds because $b_+ > 1$ while m is a small positive mortality parameter. Therefore both eigenvalues have negative real parts and b_+ is a **stable node or spiral**.

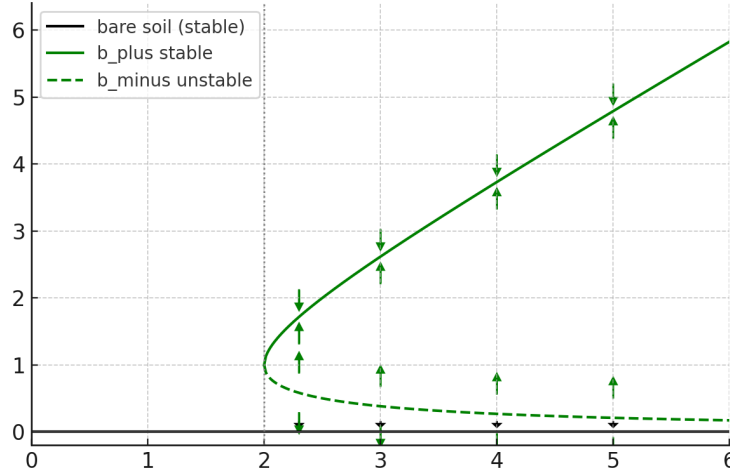


FIGURE 3. Bifurcation diagram of equilibria in the (p, b) -plane for $m = 1$.

The diagram exhibits a classical saddle–node bifurcation at the precipitation threshold $p_c = 2m$: for $p < p_c$ no vegetated equilibrium exists, while for $p > p_c$ two vegetated branches $b_-(p)$ and $b_+(p)$ appear.

- For $p < p_c = 2m$, the only equilibrium is the bare–soil state. At $p = p_c$, a saddle–node bifurcation occurs and two vegetated equilibria $b_-(p)$ and $b_+(p)$ are created.

- The lower branch $b_-(p)$ has one positive eigenvalue and is therefore unstable. (Interpretation: it acts as a biomass threshold below which vegetation collapses.)
- The upper branch $b_+(p)$ has eigenvalues with negative real parts and is stable. (Interpretation: this corresponds to a persistent vegetated state.)
- The bare-soil equilibrium remains stable for all p . (Interpretation: desert states are strongly resilient to perturbations.)
- For $p > p_c$, the system is bistable since both the bare-soil equilibrium and $b_+(p)$ are stable, separated by the unstable branch $b_-(p)$. (Interpretation: this bistability generates hysteresis in ecosystem response to precipitation.)
- The vertical arrows in the schematic indicate the sign of the biomass dynamics: trajectories move away from $b_-(p)$ and toward either the stable vegetated state $b_+(p)$ or bare soil. (Interpretation: this illustrates how initial biomass determines long-term ecosystem state.)

(c) Let $z = x - ct$. Then, we have

$$\begin{aligned} -c \frac{dw}{dz} &= p - w - wb^2 + \nu \frac{dw}{dz}, \\ -c \frac{db}{dz} &= -mb + wb^2 + \frac{d^2b}{dz^2}. \end{aligned}$$

Solving the first equation for $\frac{dw}{dz}$ gives

$$(-c - \nu) \frac{dw}{dz} = p - w - wb^2 \quad \Rightarrow \quad \frac{dw}{dz} = \frac{w + wb^2 - p}{c + \nu}.$$

By introducing $u = \frac{db}{dz}$, we can rewrite the system as

$$\begin{aligned} \frac{dw}{dz} &= \frac{w + wb^2 - p}{c + \nu}, \\ \frac{db}{dz} &= u, \\ \frac{du}{dz} &= -cu + mb - wb^2. \end{aligned}$$

Thus, the system can be expressed as a first-order system of three ODEs:

$$\frac{d}{dz} \begin{pmatrix} w \\ b \\ u \end{pmatrix} = \begin{pmatrix} \frac{w + wb^2 - p}{c + \nu} \\ u \\ -cu + mb - wb^2 \end{pmatrix}.$$

(d) The spatially uniform stationary solutions satisfy

$$\frac{\partial b}{\partial t} = \frac{\partial b}{\partial x} = \frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} = 0,$$

for $z = x - ct$, $b(x, t) = b(z)$ and $w(x, t) = w(z)$ implies

$$\frac{db}{dz} = 0, \quad \frac{dw}{dz} = 0.$$

Moreover, $u = \frac{db}{dz}$, so at such equilibria we also have $u = 0$.

Thus equilibrium solutions of the travelling-wave ODE system correspond exactly to the equilibrium of (part (b)).

Let (w^*, b^*) be an equilibrium of the original PDE system from part (b). Then, $(w^*, b^*, 0)$ is an equilibrium of the travelling-wave system.

At $(w^*, b^*, 0)$ the Jacobian of

$$F(w, b, u) = \begin{pmatrix} \frac{w + wb^2 - p}{c + \nu} \\ u \\ -cu + mb - wb^2 \end{pmatrix}$$

is

$$\begin{aligned} J &= \begin{pmatrix} \frac{\partial}{\partial w} \left(\frac{w + wb^2 - p}{c + \nu} \right) & \frac{\partial}{\partial b} \left(\frac{w + wb^2 - p}{c + \nu} \right) & \frac{\partial}{\partial u} \left(\frac{w + wb^2 - p}{c + \nu} \right) \\ \frac{\partial u}{\partial w} & \frac{\partial u}{\partial b} & \frac{\partial u}{\partial u} \\ \frac{\partial}{\partial w} (-cu + mb - wb^2) & \frac{\partial}{\partial b} (-cu + mb - wb^2) & \frac{\partial}{\partial u} (-cu + mb - wb^2) \end{pmatrix}_{(w^*, b^*, 0)} \\ &= \begin{pmatrix} \frac{1 + b^{*2}}{c + \nu} & \frac{2w^*b^*}{c + \nu} & 0 \\ 0 & 0 & 1 \\ -b^{*2} & m - 2w^*b^* & -c \end{pmatrix}. \end{aligned}$$

where we have the relation

$$w^* = \frac{m}{b^*}, \quad mb^{*2} - pb^* + m = 0,$$

from part (b), so that the Jacobian can be written as

$$J = \begin{pmatrix} \frac{1 + b^{*2}}{c + \nu} & \frac{2m}{c + \nu} & 0 \\ 0 & 0 & 1 \\ -b^{*2} & -m & -c \end{pmatrix}, \quad mb^{*2} - pb^* + m = 0.$$

The characteristic polynomial of J is given by

$$\det(J - \lambda I) = \det \begin{pmatrix} A - \lambda & B & 0 \\ 0 & -\lambda & 1 \\ -C & -m & -c - \lambda \end{pmatrix}$$

where

$$A = \frac{1 + b^{*2}}{c + \nu}, \quad B = \frac{2m}{c + \nu}, \quad C = b^{*2}.$$

Expanding the determinant along the first row:

$$\begin{aligned} 0 &= (A - \lambda) [(-\lambda)(-c - \lambda) - (1)(-m)] - B[0 - (-C)] \\ &= (A - \lambda)(\lambda^2 + c\lambda + m) - BC \\ &= A\lambda^2 + Ac\lambda + Am - \lambda^3 - c\lambda^2 - m\lambda - BC \\ &= -\lambda^3 + (A - c)\lambda^2 + (Ac - m)\lambda + (Am - BC). \end{aligned}$$

Thus, the characteristic polynomial can be written as

$$\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0,$$

where the coefficients are given by:

$$a_1 = c - A, \quad a_2 = m - Ac, \quad a_3 = BC - Am.$$

For a three-dimensional system, a Hopf bifurcation occurs when the characteristic polynomial has one real root and a pair of purely imaginary roots $\lambda = \pm i\omega$ (with $\omega > 0$). According to the Routh-Hurwitz criterion for a cubic polynomial, this occurs when:

$$a_1 > 0, \quad a_3 > 0, \quad \text{and} \quad a_1 a_2 = a_3.$$

Substituting the expressions for a_1, a_2, a_3 back into the critical condition $a_1 a_2 = a_3$:

$$(c - A)(m - Ac) = BC - Am.$$

This equation defines the relationship between the precipitation p (which determines b^* and thus A, C) and the wave speed c .

A Hopf bifurcation also requires the transversality condition, which can be checked by verifying that the real part of the critical eigenvalues crosses the imaginary axis with nonzero speed. In terms of the characteristic coefficients, this is equivalent to

$$\left. \frac{\partial}{\partial \mu} (a_1 a_2 - a_3) \right|_{\mu=\mu_c} \neq 0,$$

where μ denotes the bifurcation parameter (such as p or c).

The biomass equation shows that vegetation grows when water is available (via the term wb^2) and decreases through natural mortality ($-mb$). The water equation increases with rainfall p but decreases when vegetation consumes it (wb^2). This coupling term becomes especially important when biomass is dense, strengthening the feedback between water and vegetation.

For large precipitation p , this feedback settles to a stable, uniform vegetated state: water input balances plant uptake, and both w and b approach constant values. In this range no Hopf bifurcation occurs because all eigenvalues of the linearization have negative real parts. As p decreases toward a critical value, the balance becomes more sensitive. Dense vegetation rapidly removes water, which may reduce biomass, after which water recovers again. This feedback can create oscillations in the travelling-wave ODE system, corresponding to a Hopf bifurcation where a pair of eigenvalues crosses the imaginary axis. The resulting periodic solution in the spatial variable z represents repeating vegetation bands seen in drylands.

To look for travelling patterns, we use the same idea as for the classical wave equation

$$u_{tt} = c^2 u_{xx}, \quad u(x, t) = f(x - ct) + g(x + ct),$$

where a solution keeps its shape while moving with speed c . Introducing the travelling coordinate $z = x - ct$ gives

$$\partial_t u = -c U'(z), \quad \partial_x u = U'(z), \quad \partial_{xx} u = U''(z),$$

so the PDE becomes an ODE for the profile $U(z)$. Applying the same idea to the vegetation model,

$$w(x, t) = W(z), \quad b(x, t) = B(z),$$

reduces the PDEs to an autonomous system in z ; periodic solutions of this ODE describe spatially repeating vegetation patterns that move downslope or upslope with speed c .

The parameter c represents the slow migration speed of these bands. Because water flows downhill, the upslope edge of a vegetation band receives more runoff and expands, while the downslope edge tends to dry out and shrink. This causes the entire pattern to drift upslope even though individual plants do not move.

At the Hopf bifurcation, the linearization has eigenvalues $\lambda = \pm i\omega$. In the travelling coordinate this ω is a *spatial* frequency, so the periodic orbit has wavelength

$$L = \frac{2\pi}{\omega},$$

which gives the spacing between vegetation bands. Larger ω corresponds to more closely spaced bands, and smaller ω to wider spacing.